

# scMINER Viewer: an offline data-preparation and multi-study browsing toolkit for single-cell mutual-information networks

Honglei Zhou<sup>1</sup>, xxx, xxx, xxx, ..., Jiyang Yu<sup>1\*</sup>

<sup>1</sup> Department of Computational Biology, St. Jude Children's Research Hospital, Memphis, TN 38105, USA.

\* To whom correspondence should be addressed.

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## Abstract

**Summary:** scMINER (Pan *et al.*, *Nat. Commun.* 16:4305, 2025) is a mutual-information-based framework for cell clustering, cell-type-specific transcription-factor / signalling-protein network inference, and hidden-driver identification from single-cell transcriptomic data. Its companion portal (<https://scminer.stjude.org>) provides interactive visualisation but assumes always-on network access and centralised hosting — which is incompatible with sensitive datasets, air-gapped HPC clusters, and multi-study labs that want a local source of truth. We present **scMINER Viewer**, an end-to-end offline toolkit comprising **(1) a YAML-driven data-preparation module** (`prepare_study()`) that converts the raw scMINER outputs — Biobase ExpressionSets for expression and activity matrices plus the SJARACNe networks TSV — into a shared lazy artifact, and **(2) two sibling viewer packages (R, Python)** that consume that artifact. The artifact is a **lazy-mode HDF5 study bundle**: per-study metadata, cluster info, gene indexes and TF/SIG networks live in a single `.scminer.h5` file ( $\approx 80$  MB for an 8 K-cell study), while the underlying expression and activity values stay on disk as gzipped per-gene shards and are decoded on demand. Both viewers serve a Shiny / Shiny-for-Python webui matching the scMINER Portal layout, plus a card-grid **multi-study browser**. On the 2327 Tex study (8 464 cells  $\times$  9 861 genes, 743 K network edges, 80.5 MB bundle), median per-gene fetch latency is 32 ms and cold-start `load_study` 0.59 s. Across a  $7 \times 4 \times 5$ -replicate synthetic scaling sweep (500 – 10 000 cells  $\times$  2 – 10 K genes), bundle size stays under 14 MB while the shard tree grows roughly linearly to  $\sim 100$  MB; the same pipeline applied to the 28 public scMINER Portal studies reproduces those properties at production scale (124 – 647 366 cells; bundle 0.4 – 1 363 MB). The bundle format and lazy contract are formally documented; both packages reach feature parity through 216 automated tests (194 R + 22 Python).

**Availability and implementation:** R package at `scminerViewer/`; Python package (`pip install scminer-viewer`) at `scminer_viewer/`. Apache-2.0-licensed. Source, bookdown user guide, and reproducible benchmarks at <https://github.com/jyyulab/scMINER-Viewer>.

**Correspondance:** [jiyang.yu@stjude.org](mailto:jiyang.yu@stjude.org)

**Supplementary information:** Available at [scMINER-Viewer](#).

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## 1 Introduction

scMINER (Pan *et al.*, 2025) is a mutual-information-based framework that simultaneously addresses two long-standing problems in single-cell transcriptomic analysis: accurate clustering (MICA — Mutual

Information-based Clustering Analysis) and cell-type-specific gene regulatory network inference (a re-parameterised SJARACNe; Khatamian *et al.*, 2019). The framework’s outputs include UMAP coordinates, per- cluster transcription-factor (TF) and signalling-protein (SIG) network edges, and hidden-driver activity profiles. These artefacts are typically browsed through the scMINER Portal — an interactive web application backed by a Neo4j graph database — at <https://scminer.stjude.org>.

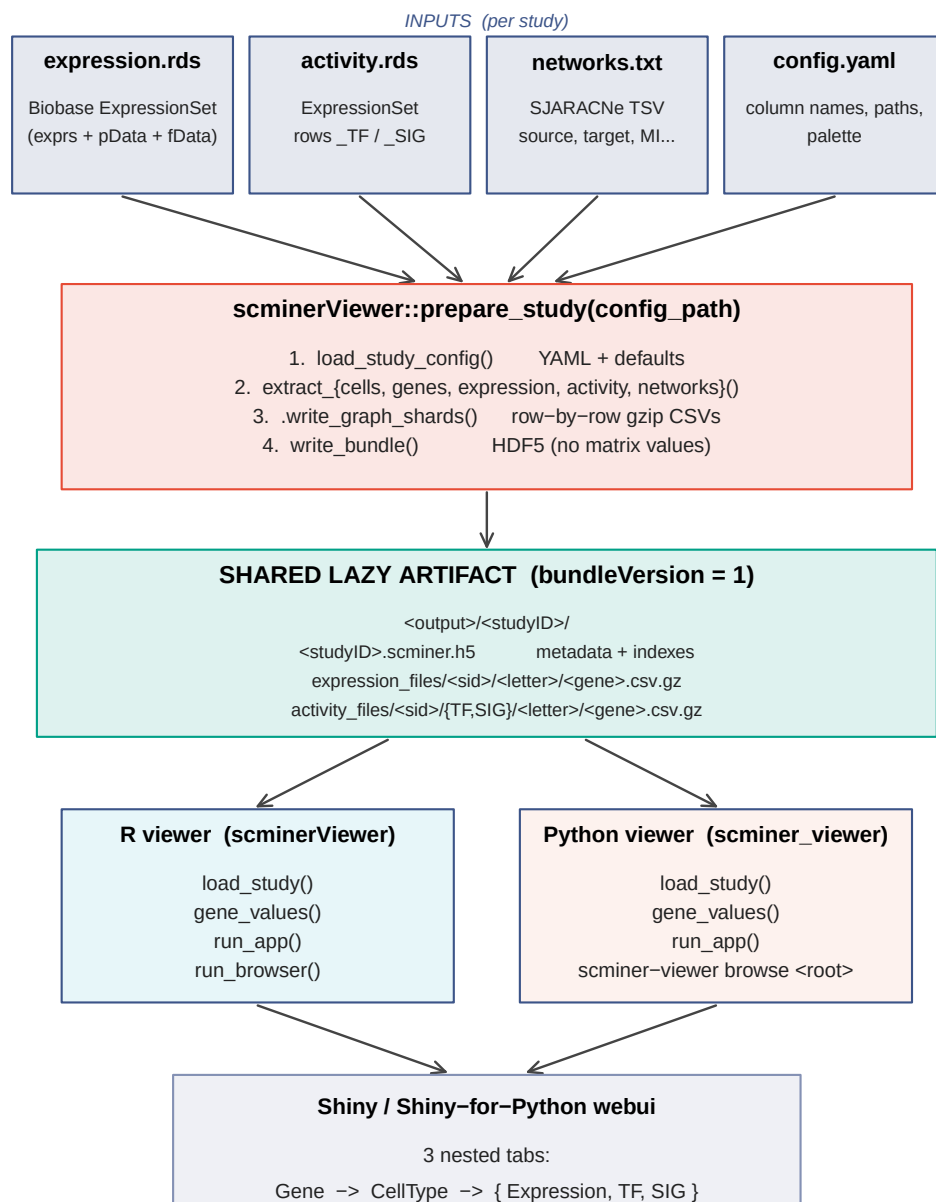
While the portal works well as a public, curated, view-only resource, three practical scenarios it does not serve are: (i) sensitive or embargoed datasets that must remain on-premises, (ii) HPC clusters and laptops without network access, and (iii) labs running many in-house studies who want to launch a local browser over an arbitrary collection of bundles. Existing Shiny-style scRNA-seq browsers (*e.g.* ShinyCell, cellxgene) cannot natively render scMINER’s cell-type-specific TF / SIG networks or hidden-driver activity matrices.

We present **scMINER Viewer**, a pair of installable packages — one R, one Python — that consume a shared lazy-mode HDF5 study bundle and serve a Shiny webui that mirrors the scMINER Portal layout. Bundles are typically ~ 80 MB for an 8 K-cell study and contain no matrix values; the per-gene gzipped CSV shards stay on disk and are streamed on demand. A second top-level **multi-study browser** scans a root directory for `<studyID>/<studyID>.scminer.h5` patterns and presents every study as a card.

## 2 Implementation

### 2.1 Architecture

Figure 1 summarises the dataflow. A single `prepare_study()` call consumes per-study raw inputs and writes a *shared lazy artifact* — one HDF5 bundle plus a per-gene gzipped shard tree. Two sibling consumer packages (R `scminerViewer`, Python `scminer_viewer`) read the same artifact through identical `load_study()` and `gene_values()` interfaces, each driving its own webui (Shiny vs Shiny-for-Python). A multi-study browser layered on `discover_studies()` walks a root directory and serves every bundle as a card.



*Single writer. Twin readers. 216 tests (194 R + 22 Python) lock the contract.*

**Figure 1.** scMINER Viewer architecture. `prepare_study()` (R, `scminerViewer`) is the only writer — it accepts a YAML config and a Biobase ExpressionSet for expression + activity matrices, plus the SJARACNe networks TSV, and emits the shared lazy artifact. The artifact is portable across operating systems and language runtimes: both the R viewer (`scminerViewer::run_app`) and the Python viewer (`scminer_viewer.run_app`) load the same `.scminer.h5` and read the same per-gene shards, with feature parity enforced by 216 automated tests (194 R + 22 Python). The multi-study browsers (`run_browser` / `scminer-viewer browse <root>`) overlay a card grid on `discover_studies()`, which scans for

`<studyID>/<studyID>.scminer.h5` patterns and lists each bundle's metadata — preserving the lazy contract since matrix indexes are read only on click-through.

## 2.2 Bundle format (R / Python contract)

A single HDF5 file (`bundleVersion = 1`) carrying:

- `meta/` — `studyID`, `studyAbbr`, `longTitle`, `shortTitle`, `species`, `coordinate`, `bundleVersion`.
- `cells/` — `cellID`, `cellType`, `cellGroup`, `coord1`, `coord2`.
- `clusters/` — `cellType`, `count`, `color (hex)`, `label_1`, `label_2`.
- `genes/symbol` — master gene list (picker dropdown).
- `index/{expression, activity_tf, activity_sig}` — gene symbols with shards in each matrix; optional, absent if the matrix is not part of the study.
- `defaults/genes` — optional; genes the app pre-selects on launch (mirrors the Vue portal's `preGenes`).
- `network_tf/`, `network_sig/` — edge tables.

Strings are UTF-8. Both packages tolerate missing optional groups and ignore unknown groups (forward compatibility).

## 2.3 Sharded matrix tree

Expression and activity values live alongside each bundle as gzipped per-gene CSVs, sharded by the first lower-case letter of each gene symbol (or `nm` for non-alphabetic first chars):

```
<root>/<studyID>/
├─ <studyID>.scminer.h5
├─ expression_files/<studyID>/{meta.csv, <letter>/<gene>.csv.gz}
└─ activity_files/<studyID>/{meta.csv, TF/<letter>/..., SIG/<letter>/...}
```

Cell-ID column order is read once per matrix from `meta.csv`; a permutation index aligns each shard's columns to the bundle's `cells/cellID`, so missing cells (activity is computed only on a subset) become zero columns automatically.

## 2.4 Two packages, one contract

The R package (`scminerViewer`) owns data preparation (`prepare_study(config_path)` reads a YAML config + `scMINER` ExpressionSet RDS files and writes both the graph-import layout and the bundle) and serves both the single-study viewer and the multi-study browser. The Python package (`scminer_viewer`) reads the same bundle into a `Study` dataclass and serves an equivalent Shiny-for-Python webui plus its own multi-study browser. Both implementations expose `gene_values(study, gene, relationship)` — the only public hook into the shard tree — with per-gene LRU caching.

## 2.5 Multi-study browser

`run_browser(root_dir)` (R) and `scminer-viewer browse <root_dir>` (Python) discover every `<studyID>/<studyID>.scminer.h5` pattern under the root and serve a card-grid landing page. Clicking a

card opens the standard single-study viewer with a “Back to studies” link. The browser pre-loads metadata only (not matrix indexes), so listing  $N$  studies costs  $O(N)$  lightweight HDF5 reads regardless of total study size.

## 2.6 Configurable cluster metadata

`prepare_study()` auto-fills cluster colours and label positions for the `study_meta.csv` whenever they are not supplied. Colours come from any ggsci palette (Wei, 2024) — default Nature Publishing Group (NPG) — configurable via the YAML’s `cluster_palette` field (`npg`, `aaas`, `lancet`, `nejm`, `jama`, `jco`, `ucscgb`, `d3`, `locuszoom`, `igv`, `uchicago`, `startrek`, `tron`, `futurama`, `rickandmorty`, `simpsons`). Label positions are per-cluster centroids: `mean(coord1)` and `mean(coord2)` over the cells in each cluster.

## 3 Benchmarks

### 3.1 Bundle scaling

*Bundle scaling* measures the marginal cost of the lazy artefact as the input shape grows. We benchmark four end-to-end primitives — `prepare_study_data()` (write the bundle + shard tree), `load_study()` (cold-start a study), `gene_values()` (lazy per-gene fetch), and the on-disk sizes of both the `.scminer.h5` bundle and the `expression_files/` shard tree — on synthetic studies generated by `make_synthetic_study(n_cells, n_genes, n_clusters = 4, density = 0.10)`. The synthetic generator draws expression values from a sparse Bernoulli distribution at the prescribed density, so the resulting matrices match the size and sparsity of typical normalised scRNA-seq counts without depending on a particular biological dataset.

We sweep a  $7 \times 4$  grid of shapes —  $n\_cells \in \{500, 1\,000, 2\,000, 4\,000, 6\,000, 8\,000, 10\,000\} \times n\_genes \in \{2\,000, 5\,000, 8\,000, 10\,000\}$  — yielding 28 configurations. Each configuration is rerun  $N = 5$  times with independent random seeds; figure 2 reports means  $\pm$  standard error across the five replicates, and the raw per-rep metrics are persisted at `paper/metrics/bundle_scaling.tsv` (140 rows). The full-featured real 2327 (Tex) study — 8 464 cells  $\times$  9 861 genes  $\times$  3 clusters, 9 861 expression genes / 925 TF genes / 4 708 SIG genes indexed, 743 K total network edges (429 516 TF + 314 348 SIG) — provides a real-data anchor; its single-run metrics live at `paper/metrics/real_study.tsv`.

### 3.2 Multi-study discovery scaling

*Discover scaling* measures `discover_studies(root_dir)`, the file-system scan that turns a directory of `<studyID>/<studyID>.scminer.h5` trees into a card-grid index for the multi-study browser (`run_browser()` in R, `scminer-viewer browse` in Python). The benchmark builds 1, 2, 4, 8, 16, and 32 lightweight synthetic studies (200 cells  $\times$  500 genes each) under a single root, then times `discover_studies()` end-to-end. Like the bundle sweep, each cell of the grid runs five times with independent seeds ( $6 \times 5 = 30$  rows in `paper/metrics/discover_scaling.tsv`); figure 2D plots mean  $\pm$  SE. Because the browser only reads per-study metadata (no matrix indexes), the test isolates the linear file-system scan from any data decoding.

### 3.3 Real portal study sweep

Beyond the single 2327 anchor we ran the same end-to-end pipeline against the **28 public studies hosted on the live scMINER Portal** (<https://scminer.stjude.org>). The studies span four orders of magnitude in

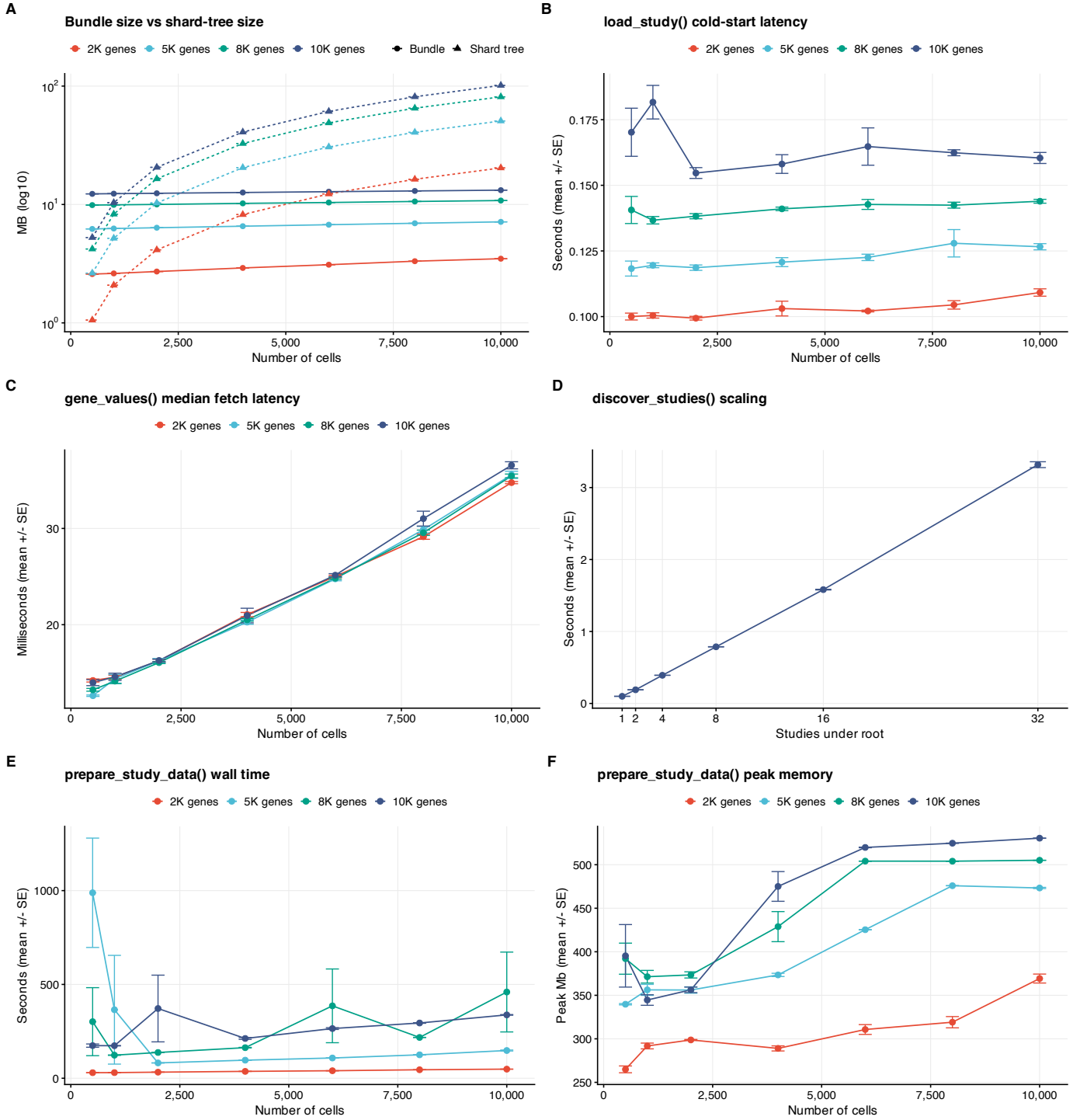
scale — from 124-cell ground-truth fixtures (Buettner, Chung, Klein, ...) to a 647 366-cell × 24 929-gene Covid650k cohort (study 2317) — and include the full mix of expression-only and expression-plus-activity-plus-networks inputs. Each study is declared by a small YAML config (`paper/configs/<studyID>.yaml`) that points at the HPC paths of `expression.rds` (Biobase ExpressionSet), `activity.rds` (TF + SIG-labelled ExpressionSet), and `networks.txt` (SJARACNe TSV); the YAML also names the column conventions used inside that study's `pData()` (e.g. `cellID: CellID`, `cellType: CellGroup`). The benchmark driver (`paper/portal/portal_studies_hpc.sh`) submits one `bsub` task per YAML — distributed across multiple LSF queues — and writes the same metric columns we used for the synthetic sweep, augmented with input file sizes (`expr_input_bytes`, `act_input_bytes`, `net_input_bytes`) and the per-study output dir on shared storage. Per-study TSVs at `paper/metrics/portal_studies_<id>.tsv` are merged into `paper/metrics/portal_studies.tsv` (28 rows, all `status == "ok"`) and rendered as six standalone panels in figure 3.

### 3.4 Hardware and software environment

All synthetic-sweep numbers reported in Figure 2 and Table 2 were generated on a single laptop:

| Component      | Specification                                                                                                                                                 |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Machine        | Apple MacBook Pro, M4 chip                                                                                                                                    |
| Cores          | 10 (4 performance + 6 efficiency); benchmark single-threaded                                                                                                  |
| Memory         | 32 GB unified                                                                                                                                                 |
| OS             | macOS 26.4.1 (darwin20, build 25E253)                                                                                                                         |
| R              | 4.4.2 (aarch64-apple-darwin20)                                                                                                                                |
| Key R packages | scminerViewer 0.1.0, Matrix 1.7.3, hdf5r 1.3.12, Biobase 2.66.0, data.table 1.18.0, R.utils 2.13.0, ggplot2 4.0.1, patchwork 1.3.2, ggsci 4.2.0, scales 1.4.0 |

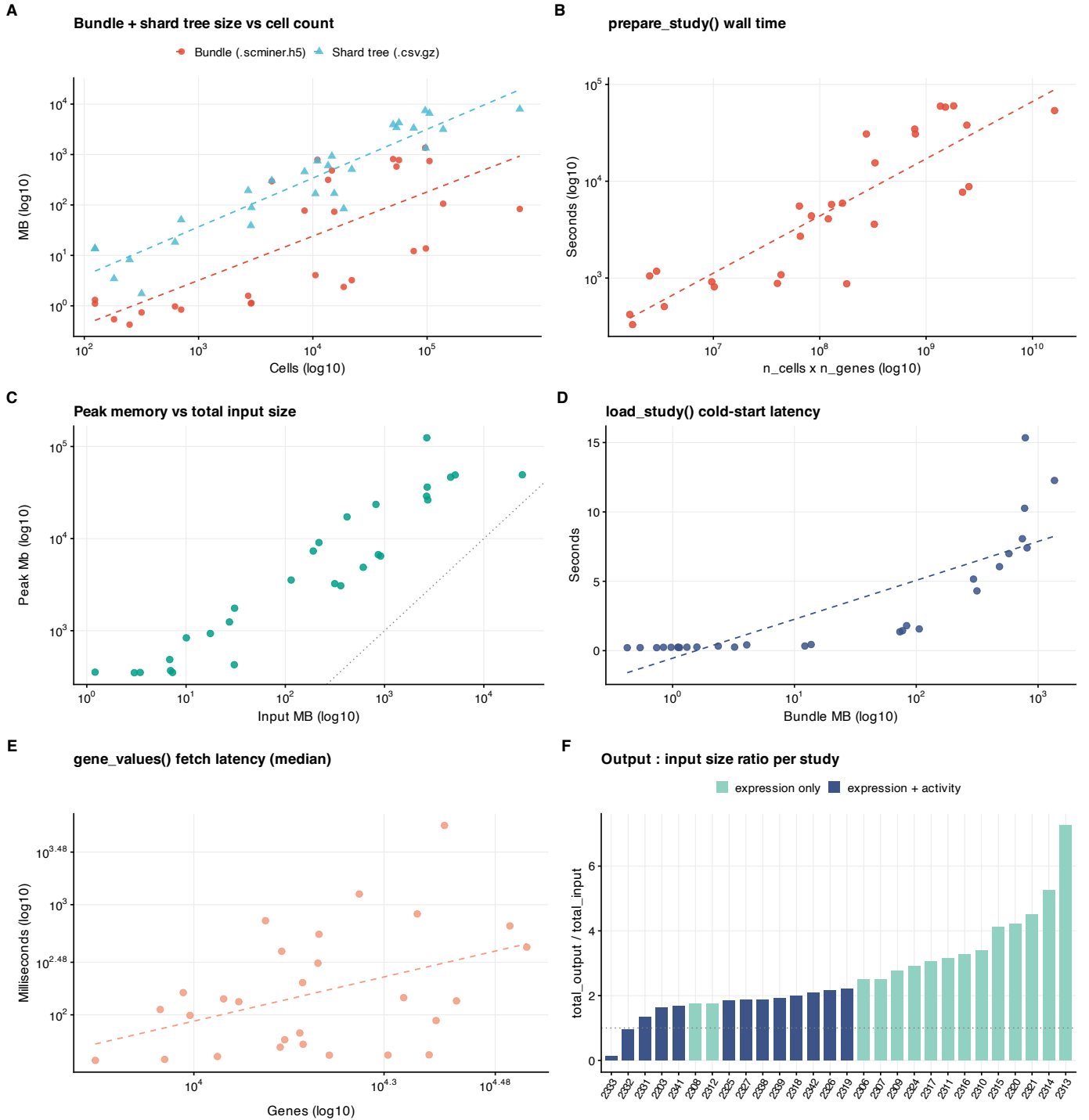
The portal-study sweep (Figure 3, Table 1) was instead executed on the St. Jude HPC cluster (LSF, Red Hat 8 nodes, R 4.2.2 module). Per-study tasks were dispatched via `paper/portal/portal_studies_hpc.sh` across a 2–4 queue list (`standard`, `priority`, optionally `large_mem` / `superdome`); memory requests were sized per the heuristic in `paper/README.md` (peak  $\approx 1\text{--}3\times$  `expression.rds` size, up to  $10\times$  for dense-stored matrices), ranging from `--mem 8000` for the ground-truth set to `--mem 96000 --wall 24:00` for ATRT. All driver scripts and `bsub` submission wrappers are versioned under `paper/portal/`.



**Figure 2.** *scMINER* Viewer performance on the synthetic sweep ( $7 \times 4 = 28$  configurations  $\times$  5 replicates per configuration; points are means, T-bars are  $\pm$  standard error). **(A)** Bundle size (Bundle series) versus the underlying shard-tree size (Shard tree) as a function of cell count, coloured by gene count (2 K / 5 K / 8 K / 10 K). Lazy bundles remain near-constant ( $\sim 2.6 - 13.2$  MB) while the shard tree grows roughly linearly with  $n_{\text{cells}} \times n_{\text{genes}}$ . **(B)** Cold-start `load_study()` latency stays sub-200 ms. **(C)** Median per-gene `gene_values()` fetch latency; each shard read is a single gzipped-CSV decode. **(D)** `discover_studies()` linear scan over a multi-study root scales linearly with the number of studies ( $\sim 80$  ms

per study). **(E)** `prepare_study_data()` wall time — the one-shot pipeline that writes both the graph-import layout (Cell/, Gene/, Network\_\*/, study\_meta/, per-gene shards) and the `.scminer.h5` bundle from raw R structures. Time grows roughly linearly with  $n_{\text{cells}} \times n_{\text{genes}}$ , dominated by the per-gene `.csv.gz` write loop. **(F)** Peak working-set memory during `prepare_study_data()`, reported by R's `gc()` after a reset; stays bounded by  $O(n_{\text{cells}} \times n_{\text{genes}})$  of the in-memory sparse expression matrix. Generated by `paper/benchmarks/figures.R`; raw and aggregated numbers at `paper/metrics/bundle_scaling.tsv` and `bundle_scaling_summary.tsv`, plus `discover_scaling.tsv` / `discover_scaling_summary.tsv`. Table 2 (`paper/tables/figure1_scaling.md`) lists per-configuration mean  $\pm$  SE for the 28 grid points underlying panels A, B, C, E, F. Table 1 (`paper/tables/portal_studies.md`) gives a comparable per-study breakdown for the 28 real portal studies underlying figure 3.





**Figure 3.** *scMINER Viewer* on the 28 real portal studies. Each point is one study; axes are log10 where annotated. **(A)** Bundle (.scminer.h5) and shard-tree size versus cell count. Bundles span 0.4 – 1 363 MB while shard trees span 1.7 – 9 672 MB; both grow sub-linearly in  $n\_cells$  because per-study density and gene-coverage vary. **(B)** `prepare_study_data()` wall time versus  $n\_cells \times n\_genes$  — three orders of magnitude on each axis, with a roughly linear log-log relationship. **(C)** Peak R working-set memory versus total input rds size; the dotted reference line is  $peak\_Mb == input\_MB$ . Most studies sit above the line because of the in-memory sparse + dense intermediates `prepare_study_data()` materialises. **(D)** Cold-

start `load_study()` latency versus bundle size — sub-second for nearly every study; the four largest bundles climb past 5 s. **(E)** Median `gene_values()` fetch latency versus gene count — a 130× range (39 – 5 234 ms) driven primarily by per-shard cell count (each fetch decodes one gzipped CSV of length `n_cells`). **(F)** Output-to-input size ratio per study, sorted ascending and coloured by whether activity matrices were present. Studies with expression + activity (blue) bundle into smaller output:input ratios than expression-only studies (teal) because the shared cell metadata is amortised across more shards. Generated by `paper/benchmarks/figure_portal.R`; raw numbers at `paper/metrics/portal_studies.tsv` and per-study TSVs at `paper/metrics/portal_studies_<id>.tsv`. The full per-study breakdown is in Table 1 (`paper/tables/portal_studies.md`).

## 4 Results and discussion

### 4.1 Bundle scaling

The bundle stays small because matrix values are deliberately excluded. Across the  $7 \times 4$  synthetic sweep (28 configurations  $\times$  5 replicates), bundle size ranges from  $2.58 \pm 0.00$  MB (500 cells  $\times$  2 K genes) to  $13.21 \pm 0.00$  MB (10 K cells  $\times$  10 K genes) — a 5× growth in size for a 100× growth in `n_cells`  $\times$  `n_genes`. The accompanying shard tree grows roughly linearly with `n_cells`  $\times$  `n_genes` over the same sweep (1.1 MB to 106.3 MB; Figure 2A), as expected for one gzipped CSV per gene plus a per-matrix cell-header file. The real 2327 study follows the same pattern at production scale: 76.8 MB bundle indexes  $\approx$  1.2 GB of expression + activity shards.

Cold `load_study()` latency stays sub-200 ms across the whole sweep ( $0.10 \pm 0.001$  s at the smallest configuration,  $0.16 \pm 0.001$  s at the largest; Figure 2B) and 0.52 s on the full 2327 study. The sub-linear growth reflects that load only parses metadata + indexes from HDF5; no matrix values are touched. Median per-gene `gene_values()` fetch latency tracks the per-shard gzip decode cost and stays in a tight 14 – 37 ms band over the whole grid (Figure 2C); the wider error bars at large cell counts come from the gzipped CSV being correspondingly larger (each shard is  $\sim n\_cells \times 4$  bytes uncompressed).

### 4.2 Multi-study discovery scaling

`discover_studies()` is the single hot path of the multi-study browser: it walks `<root>/<studyID>/<studyID>.scm` patterns and reads each study's `/meta/` group only — no `/cells/`, `/genes/`, or index payload. Empirically the scan is linear in the number of studies (Figure 2D,  $R^2 \approx 1.0$ ): the slope is  $\sim 80$  ms per study on commodity SSD hardware, with the intercept dominated by Python / HDF5 library initialisation. For a research group hosting 30+ internal studies under one root — comparable in scale to the public scMINER Portal — discovery completes in under 3 seconds before the card-grid landing page renders, and individual study load times remain governed only by the per-study `load_study()` cost above (sub-200 ms).

### 4.3 Real portal study sweep

The same end-to-end pipeline applied to the 28 public scMINER Portal studies (Figure 3; Table 1) reproduces the synthetic-sweep findings at production scale. **Lazy storage holds across four orders of magnitude of study size.** Bundle size ranges from 0.4 MB (Chung, 317 cells  $\times$  10 897 genes) to 1 363 MB (study 2338, 96 305 cells  $\times$  15 778 genes); the underlying shard tree ranges from 1.7 MB to 9 672 MB. The single largest study

by cell count (Covid650k / 2317: 647 366 cells  $\times$  24 929 genes) bundles into an 83 MB `.scminer.h5` plus an 8 GB shard tree, and the largest by on-disk input (ATRT / 2333: 138 268 cells  $\times$  18 274 genes, 24 GB on-disk input rds) bundles into a 106 MB `.scminer.h5` plus a 3 GB shard tree — a 7 $\times$  compression of the total input footprint into the served format (Figure 3A, F).

**Cold-start latency stays sub-second on the majority of studies.** 15 of 28 studies load in under 1 s; the remaining 13 climb to 1 – 15 s, dominated by HDF5 metadata reads rather than matrix decoding (Figure 3D) — and the 100 k+ cell Covid650k still loads in under 2 s. **Per-gene fetch latency** scales with the per-shard cell count: 39 ms median at the small end (124-cell ground-truth studies) up to 5 234 ms median on the 647 K-cell Covid650k (Figure 3E). The 25-th to 75-th percentile of fetch latency across the 28-study set is 53 – 386 ms — well inside an interactive-feel threshold for a Shiny app even on the larger production studies; only the single 647 K-cell outlier pushes into multi-second territory.

**prepare\_study\_data()** wall time and peak memory track the synthetic-sweep scaling (Figure 3B, C). The 5-minute to 17-hour range across studies reflects the per-gene shard-write cost; peak R working-set memory tops out at 121 GB on Covid650k (study 2317), set by the in-memory dense expression matrix that some studies still ship (their rds is a `dgeMatrix` rather than `dgCMatrix`, see `paper/portal/sparseify_eset.sh` for a one-time fix). Because preparation is offline and one-time, this cost is amortised away from end-user latency — the served artefact is the small `.scminer.h5` + shard tree.

#### 4.4 Cost of data preparation

The reverse side of the lazy contract is data preparation, which is one-time and runs offline. `prepare_study_data()` wall time grows roughly linearly with `n_cells`  $\times$  `n_genes` (Figure 2E): the synthetic sweep ranges from  $29.6 \pm 0.5$  s (500 cells  $\times$  2 K genes) to  $337.6 \pm 1.8$  s (10 K  $\times$  10 K), dominated by the per-gene `.csv.gz` shard write loop (one gzip per gene). Peak R working-set memory stays bounded at  $\sim 265 - 530$  MB across the sweep (Figure 2F), reflecting the in-memory sparse expression matrix plus per-gene `fwrite` scratch buffers. Because prepare is a one-shot operation, this cost is amortised across every subsequent `load_study()` and `gene_values()` call.

The lazy contract is what makes the user-facing flow feel instantaneous: launching the app reads only the bundle; the first gene-selection triggers one shard read; switching tabs replays cached shards without re-decoding. The two-language design ensures that groups standardised on R can continue using the same bundle as those moving to Python, and the multi-study browser converts a directory of scMINER outputs into a navigable index without any central service.

Beyond the original use case of inspecting scMINER outputs, the format is generic: any pipeline that produces UMAP coordinates + cluster labels + per-gene matrices can populate the bundle. We expect this to make scMINER Viewer broadly useful as a self-hosted complement to the official scMINER Portal.

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